

AI Concepts for Official Statisticians

Learner guide

OS-A02-A03 | Version 1.0 | 26 September 2026

Purpose

This guide helps you classify an AI-assisted statistical task, identify the evidence and controls it requires, and issue a defensible decision. It converts the accepted canonical lesson into procedures, worked examples, checklists and a workplace transfer route.

Use this guide to	Evidence you produce
Prepare	Baseline classification and uncertainty notes.
Practise	Task map, mechanism statement and failure register.
Perform	AI Task and Technology Classification Dossier.
Transfer	Thirty-day plan for one authorized workplace task.

Professional boundary

Use only public, synthetic, de-identified or formally approved controlled material. Never enter confidential, identifiable, embargoed or security-sensitive information into an unapproved service. AI may assist analysis; it does not acquire methodological, professional or release authority.

Learning outcomes and route

Outcome	You can demonstrate it by
LO1 Task decomposition	Separating judgement, information, transformation, prediction, generation, optimization and control.
LO2 Mechanism classification	Distinguishing rules, statistical models, machine learning, deep learning, generative AI, retrieval and agents.
LO3 Assumptions	Stating inputs, outputs, objective, evidence, uncertainty and human boundary.
LO4 Failure analysis	Tracing leakage, bias, drift, hallucination, retrieval failure and compounding error.
LO5 Official statistics	Connecting failures to quality, confidentiality, reproducibility and authority.
LO6 Decision	Issuing permit, redesign, reject or escalate with residual uncertainty.

Study route

- Read the mechanism pages before selecting a technology label.
- Complete the seven-question worksheet for every case.
- Compare the candidate with the current or simpler baseline.
- Record uncertainty and evidence gaps; do not manufacture certainty.
- Use the dossier and checklist before submitting the terminal task.

Success standard

Competent performance normally requires at least 70 of 100, no critical failure, and at least half the available score in every mandatory criterion. Lead-level work reaches at least 85, challenges the evidence, compares credible alternatives and defends a decision under uncertainty.

The seven-question method

Step	Write down	Do not accept
1 Purpose	Product, user, population, consequence, excluded use.	“Use AI to improve efficiency.”
2 Work	Tasks, dependencies, judgement, baseline and exceptions.	A single undifferentiated workflow.
3 Mechanism	What is encoded, learned, retrieved, generated or optimized.	A vendor or marketing label.
4 Evidence	Reference standard, evaluation population, metrics, uncertainty and tolerances.	Training fit or anecdotes.
5 Failure	Mechanism, affected group and propagation into the product.	A generic risk list.
6 Control	Verification, correction, stopping, fallback and authority.	“Human in the loop” without capacity or power.
7 Decision	Permit, redesign, reject or escalate; conditions and review date.	Approval based on plausibility.

Decision meanings

- Permit: bounded experiment supported by proportionate evidence and controls.
- Redesign: plausible value, but evidence, workflow or controls are inadequate.
- Reject: a mandatory gate fails or residual risk is disproportionate.
- Escalate: law, rights, public value or release authority exceeds your mandate.

Mechanism classification guide

Family	Diagnostic question	Characteristic evidence
Rules	Are outputs produced by declared conditions?	Specification conformance and exception tests.
Statistical model	Is a stochastic relationship estimated with interpretable assumptions?	Model diagnostics, inference and population conditions.
Machine learning	Is a predictive function fitted from examples?	Independent generalization and class-specific error.
Deep learning	Are layered representations learned?	Domain transfer, perturbation and run stability.
Generative AI	Is new content sampled conditionally?	Claim support, variability, correction and refusal.
Retrieval system	Are sources selected and assembled before an answer?	Corpus coverage, recall, authority and citation support.
Optimization	Are actions selected against an objective and constraints?	Objective validity, constraints and sensitivity.
Agentic workflow	Does a system plan and use tools across steps?	Permissions, checkpoints, composition tests and recovery.

Mechanism statement template

The proposed system takes [inputs], uses [encoded, learned, retrieved, generated or optimized function] to produce [outputs] for [bounded task]. It may adapt through [mechanism]. It is not authorized to [excluded decisions]. Human officials retain [judgements and authority].

Task decomposition and baseline

Start with the statistical workflow, not the proposed technology. A component can perform well on clean inputs while failing in the composed production process.

Task field	Learner entry
Purpose and official product	What must be produced, for whom and under what mandate?
Inputs and provenance	Which sources, versions, populations and access conditions?
Transformation	Which edits, links, classifications, estimates or narratives?
Professional judgement	Where are concepts, thresholds, interpretation or release decided?
Current baseline	Quality, time, burden, cost, known failures and exceptions.
Candidate position	Exactly which bounded task changes?
Handoffs	What enters and leaves the component?
Fallback	How does work continue if the candidate is unavailable or stopped?

Quality-adjusted time

Count preparation, generation, verification, correction, exception handling, documentation and recovery. Faster initial output is not an efficiency gain when checking burden, error or institutional dependence increases.

Exercise

Decompose occupation coding into capture, normalization, candidate generation, consistency checking, adjudication, weighting, quality measurement and release. Mark which judgements remain non-delegable.

Evidence and evaluation

Question	Minimum evidence
For whom?	Declared evaluation population and sampling mechanism.
Compared with what?	Credible current or simpler baseline under comparable conditions.
Correct according to what?	Independent reference standard and adjudication protocol.
Measured how?	Metrics tied to error consequences, with uncertainty.
Works where?	Domain, geography, language, subgroup and time analysis.
After integration?	End-to-end workflow, reviewer capacity and product consequence.
For how long?	Monitoring, drift evidence, expiry and revalidation triggers.

Uncertainty ledger

- Sampling: the evaluation sample may not represent use.
- Coverage and selection: units or labels may be missing systematically.
- Measurement and construct: proxies may not represent the intended concept.
- Model: estimates vary and may fail after distribution change.
- Retrieval and generation: sources may be absent; claims may be unsupported.
- Decision: thresholds reflect consequences and values, not performance alone.

Evidence warning

Accuracy, model confidence, token probability and citation appearance answer different questions. None alone establishes that an official claim is valid.

Failure analysis and quality mapping

Failure	Propagation question	Official-statistics concern
Specification error	Does the system execute an invalid rule correctly?	Accuracy, coherence, authority.
Leakage or selection	Did development evidence use unavailable or unrepresentative information?	Validity, coverage, impartiality.
Distribution shift	Have population, questionnaire, source or classification changed?	Comparability, reliability.
Hallucination	Is a fluent claim unsupported or contradictory?	Accuracy, clarity, trust.
Retrieval failure	Was the controlling source absent, outdated or misranked?	Accessibility, authority.
Subgroup harm	Are error or abstention burdens unequal?	Public value, impartiality.
Automation bias	Does confidence reduce independent human correction?	Professional independence.
Compounding error	Do multi-step failures amplify or alter later data?	Auditability, continuity.

Failure statement template

Because [mechanism or condition], [error] may occur for [population or domain]. It can propagate through [workflow path] and affect [official product, decision or group]. Detect it using [evidence]. Contain it through [control and fallback]. Residual uncertainty is [statement].

Worked case occupation coding

Kintaba has 40,000 historical coded descriptions and 600,000 uncoded records. The manual baseline achieves 91.2 percent adjudicated accuracy at 4.8 minutes per record. A supervised candidate reaches 94.6 percent overall, but recall is 81.3 percent for informal-work descriptions and 76.8 percent for the two smallest language strata. The pre-declared tolerance is at least 90 percent recall in every stratum and no increase in corrected classification error.

Method step	Worked reasoning
Purpose	Assist candidate-code generation; do not automate classification authority.
Mechanism	Supervised classifier trained on historically coded descriptions.
Evidence problem	Historical labels encode inconsistent treatment; aggregate accuracy hides sparse strata.
Failure consequence	Misclassification affects occupational distributions and shifts workload to minority-language coders.
Controls	Adjudicated stratified reference sample, manual review, thresholding, random QC and drift monitoring.
Decision	REDESIGN: repair labels and subgroup evidence; retain manual route for low-support strata.
Residual uncertainty	Emerging occupations and language change require rotating reference samples.

Your challenge

Explain why 94.6 percent overall accuracy cannot justify autonomous coding. Then propose one alternative baseline and one stopping trigger.

Four additional Kintaba decisions

Case	Evidence	Defensible boundary
Record linkage	Learned matcher reduces missed links to 4.1% but raises false links to 2.3%; 5.9% for recent migrants.	Permit sandbox analysis with sensitivity testing; reject operational register updating.
Crop area	Pixel accuracy is 92.7%, but area is understated 11.8% in intercropped zones and 15.4% for fragmented plots.	Redesign around calibrated estimation and ground surveys; exclude unsupported domains.
Release drafting	11-minute draft plus 35-minute verification versus 52-minute baseline; material corrections rise from 0.7 to 1.9.	Redesign for public inputs and traceable drafts; no autonomous interpretation or release.
Public query support	Controlling-source recall is 93.8% overall, 82.0% for minority-language queries; refusal 88.6% versus 98% tolerance.	Redesign retrieval and refusal; retain answer, qualify, refuse or refer routes.

Comparative reflection

- Which component metric improves while the official product remains at risk?
- Which group or domain carries the residual error?
- Which boundary is non-compensatory?
- What new evidence could change the decision?

Responsible AI-assisted work procedure

Phase	AI may assist	You must retain
Frame	Organize public requirements and candidate task maps.	Purpose, construct, population, mandate and exclusions.
Explore	Summarize approved sources and suggest alternatives.	Source authority, completeness and relevance.
Build	Draft code, prompts, tests or candidate classifications.	Specification, secure environment and test design.
Evaluate	Calculate metrics and compare declared alternatives.	Reference standard, tolerances, uncertainty and interpretation.
Verify	Flag inconsistencies and assemble evidence records.	Claim-level checking, correction and exception decisions.
Decide	Structure comparison and draft a decision record.	Professional judgement, stopping, escalation and release authority.

Prompt record

Record the task, approved context, requested output, exclusions, model or service version, retrieval sources, parameters, response, verification, corrections and final human decision. A prompt is a controlled input, not evidence of validity.

Safe instruction pattern

Using only the approved material below, produce [bounded output]. Distinguish source statements from inference. Cite each material claim to the supplied source. State missing evidence. Do not infer confidential detail or make release decisions. Return [structured format] for human verification.

Terminal dossier template

Section	Required content	Assessment focus
1 Purpose and boundary	Need, product, user, population, consequence and exclusions.	10 points
2 Task map	Steps, dependencies, judgement, current baseline and handoffs.	15 points
3 Technology classification	Mechanism, inputs, output, objective and adaptation.	15 points
4 Evidence and assumptions	Reference standard, population, metrics, uncertainty and tolerances.	15 points
5 Failure register	Statistical, technical and human failures with propagation.	20 points
6 Controls and authority	Verification, access, stopping, fallback, escalation and release.	15 points
7 Decision record	Alternatives, decision, conditions, residual uncertainty and review date.	10 points

Critical failures

- Automating or authorizing a decision outside the mandate.
- Concealing consequential judgement inside an automated step.
- Fabricating evidence or claiming validation not performed.
- Omitting a known failure that could materially invalidate or harm use.
- Using protected information in an unapproved service.
- Assigning statistical or release authority to the system.
- Approving solely on plausibility, supplier assurance or aggregate score.

Verification checklist and workplace transfer

Check	Evidence present
Purpose and exclusions	☐
Current and simpler baselines	☐
Mechanism stated without anthropomorphism	☐
Independent reference evidence	☐
Uncertainty and pre-declared tolerances	☐
Subgroup and domain performance	☐
Confidentiality and supplier boundary	☐
Verification and correction burden	☐
Stopping, fallback and restart rules	☐
Named authority and release owner	☐
Residual uncertainty and review date	☐

Thirty-day transfer

Period	Action
Days 1 to 5	Select one authorized bounded task and document the baseline.
Days 6 to 10	Decompose work and classify the mechanism.
Days 11 to 15	Specify evidence, uncertainty, tolerances and affected groups.
Days 16 to 20	Map failures, confidentiality, controls and authority.
Days 21 to 25	Compare alternatives and draft the decision record.
Days 26 to 30	Obtain supervisor review; revise and record next evidence event.

Transfer boundary

This exercise does not authorize production deployment. Use approved material and institutional processes. Escalate unresolved legal, confidentiality, rights, public-value or release questions.

Glossary and controlled references

Term	Meaning in this module
Baseline	Current or simpler alternative measured comparably.
Calibration	Agreement between stated probability and observed frequency.
Concept drift	Change in the relationship between inputs and target.
Grounding	Constraining output through identified source context.
Hallucination	Unsupported or contradictory generated content.
Quality-adjusted time	Total effort including verification, correction and recovery.
Reference standard	Evidence used to judge correctness.
Residual uncertainty	Material uncertainty remaining after controls.
Stopping rule	Observable trigger, owner, containment, fallback and restart condition.
Total error	Combined errors across coverage, measurement, processing, estimation and use.

References

- United Nations. Fundamental Principles of Official Statistics, 2014.
- UNSD. National Quality Assurance Frameworks Manual, 2019.
- UNECE. Machine Learning for Official Statistics, 2022.
- UNECE. Responsible AI for Official Statistics Framework, 2025.
- NIST. AI Risk Management Framework 1.0, 2023.
- OECD. Updated Definition of an AI System, 2024.
- SRAI. OS-A02-A01 Canonical Lesson v0.2, accepted 26 September 2026.